



Figure 1: Various aspects of scholarly academic research

the first step in this process is searching in specialised search engines such as Google Scholar and Microsoft Academic. The crucial component is how to collect and organise the results returned by these search engines in response to the user query. To carry out this step, various tools are available, from which Zotero has been selected.

Zotero is powerful software for collecting and managing references for your research. It can be installed in two different modes:

- Standalone
- Firefox Extension
- Zotero provides five major features to users:
 - Collect
 - Organise
 - Cite
 - Sync
 - Collaborate

The Zotero workflow is as follows.

- You navigate to the *Results* page of an academic search engine or any page that contains your research content.
- Zotero comes loaded with various components that sense the presence of research content on a page. If sensed, automatically an icon with the Zotero logo is shown in the browser. By simply clicking on this icon, the metadata associated with that content is automatically added to your Zotero repository.
- While you are writing a paper, you can directly insert this reference, by searching in the Zotero word processor connector plugins. These plugins can be installed either manually or automatically.

The biggest advantage of Zotero is that changing the format of the citation style to another can be done with just a single click. For scholarly researchers, this task of changing the layout and citation style of a manuscript from one format to another is a laborious task. For example, if you prepare a manuscript with the IEEE citation style and now wish to convert that to Springer style, then you can do so by simply changing the *Citation style* settings in the Zotero word processor plugin. The Zotero style library has more than 8100 different styles. Zotero allows the creation of custom styles as well.

Zotero provides the cloud syncing capability, which allows you to sync your collected research data across various systems. There is a detailed article about Zotero available at



Figure 2: Major features of Zotero

<http://opensourceforu.com/2015/01/speed-up-your-research-paper-with-zotero/>

Drafting a manuscript

Drafting a manuscript is another big task. Though any word processor can be used to draft manuscripts, there are some inherent limitations in this approach. As stated earlier, converting a manuscript from the style template of one journal to another is a task that cannot be automated if you simply go with the word processor approach.

To rescue scholars from the burden of such laborious tasks, we have **LaTeX** (usually pronounced as Lay-Tech). It is a typesetting system for creating production quality output. While the word processing tools follow the 'What You See is What You Get' (WYSWYG) approach, LaTeX uses an approach similar to markup languages such as HTML. Here, a clear demarcation is made between the content and its style. The authors can concentrate more on the content. Production of professional quality output, as per the style selected by the author, is carried out by the LaTeX software. It is obvious that getting familiar with LaTeX requires some initial extra work by the authors. However, it is worth going through this learning curve, as the benefits are huge.

Shown below is a simple LaTeX example:

```

\documentclass{article}
\title{Simple LaTeX Article}
\author{K S Kuppusamy}
\date{01 Jan 2018}
\begin{document}
\maketitle
Hello world!
\end{document}
  
```

The meaning of this LaTeX code is given below:

- This LaTeX document belongs to the class called 'Article'. There are other classes such as 'Report'.
- The title for this article is 'Sample LaTeX Article'.
- The author's name is 'K S Kuppusamy'.
- The date of this article is '01 Jan 2018'.
- The main body content is 'Hello world!'.